

Wondering what MOOCs are?

These are basically classes without walls, which can accommodate billions of students from around the world. Watch this TED talk by Prof. Anant Agarwal (www.youtube.com/watch?v=eW3gMGqc2Dc) to learn more about this trend.

Virtual labs

- CircuitLogix 3D Lab (www.circuitlogix.com/circuitlogix-3DLab.php)
- DC/AC Virtual Lab (www.dcaclab.com/)
- DoCircuits (www.docircuits.com/)
- IIT Bombay e-Prayog (<http://59.181.142.81/>)
- IIT Kharagpur Digital Electronics Circuits Virtual Lab (www.digital.iitkgp.ernet.in/dec/)
- TECH FOCUS: Real Electronics, Learnt Virtually Virtual Labs (www.vlab.co.in/)
- University of Colorado's Circuit Construction Kit and Virtual Lab (<http://phet.colorado.edu/en/simulation/circuit-construction-kit-ac-virtual-lab>)

Webinars, communities and more

- EDA Board (www.edaboard.com/)
- Electronic Design (<http://electronicdesign.com/learning-resources/webcasts>)
- Electronics For You (<http://forum.electronicsforu.com/>)
- EFY's Electronic Design Community (www.facebook.com/designelectronics)
- EEVblog Electronics Community Forum (www.eevblog.com/forum/)
- Electronics Lab Community (www.electronics-lab.com/forum/index.php)
- EEWeb (www.eeweb.com/electronics-forum)
- IEEE Spectrum (<http://spectrum.ieee.org/webinars>)

Virtual classes

- Class Central (www.class-central.com/)
- Coursera (www.coursera.org/)
- EdX (www.edx.org)
- Khan Academy (www.khanacademy.org/)
- The Institution of Engineering and Technology (www.theiet.tv)
- Udemy (www.udemy.com/)

MOOC repositories to break barriers and let everybody learn. You may check out MIT's RSS feed (<http://ocw.mit.edu/help/ress/>) and Carnegie Mellon's free courses (<http://oli.cmu.edu/>), or if you are keen on learning about artificial intelligence (AI), then Stanford Engineering Everywhere (<http://see.stanford.edu/>) comes highly recommended.

Udemy is another wonderful website that not only allows you to learn but also build courses for others to learn from. It has over 25,000 courses and six million students. So if you are an expert in any stream and wish to teach the same to others, do your bit and upload a free course.

Experiment to your heart's content

Labs are an integral part of learning, especially when it comes to engineering and vocational education.

Virtual Labs, an initiative by Ministry of Human Resource Development under National

Mission on Education through ICT is a very appreciable indigenous initiative that aims to provide remote access to virtual lab facilities to schools and other vocational trainers, so that everybody, in any part of the country, gets access to advanced learning tools.

While a number of labs in varied streams are available, some of those pertaining to electronics and communications include interactive design and test, microwave, wireless, electromagnetic, queuing networks modelling, advanced communications, hybrid electronics, transducer and instrumentation, very large scale integration (VLSI) design and verification, microcontrollers, signal

processing, speech signal processing, fibre optics and other labs relevant to current-generation electronics education.

Several IITs, Amrita University, IIT Hyderabad and other institutions are contributing their resources and efforts to Virtual Labs project on an ongoing basis. You can register for free with IIT Bombay's e-Prayog or IIT Kharagpur's Digital Electronics Circuits Virtual Lab to try your hand at experiments that reinforce your theoretical learning.

Take some quick lessons and get your doubts answered

When you do not have time for a full course and just need a quick how-to, overview or jargon teardown, you have many options, notably articles, webinars and forums. In many cases, a webinar from a reliable source is more effective than reading a dozen articles because these are usually conducted by experienced people who know what you need to know.

It is also important that you be a part of online communities and forums. This will help you to understand the latest trends as well as get your questions answered, whenever required. Remember that you can get your doubts cleared only if you are active on the forum, and help others by sharing your knowledge too. Forums and online communities will also help you find collaborators for your projects, if needed.

While we have mentioned some general forums and communities in the box above, it is advisable to join forums and communities specific to your areas of interest. This way you can gain advanced knowledge and in-depth responses.

Whether you wish to learn the basics, acquire a new skill, develop a product or even start a company, resources are aplenty today. Just ensure that you are completely dedicated to your undertaking and are willing to put in the required time and effort. You will surely taste the sweet fruit of success. ●